SCX Audit of Governance:

Game-Theoretic Foundations of Transparency Under Multi-Expert Verification
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Abstract

We formalize governance as a signaling game with multi-expert audit under the SCX (Structured Causal eXamination) framework. A government G possesses private information about the true state of society θ—including GDP growth rate, unemployment, fertility rate, and pollution levels—and publishes a claim m. An auditor community $\mathcal{A} = \{A_1, \ldots, A_M\}$ independently estimates θ from heterogeneous data sources (household surveys, satellite imagery, administrative records, third-party data). The government's payoff follows the YAJIE Nash-Pareto Equilibrium structure: benefit from favorable reporting minus expected cost when the claim deviates from multi-expert consensus. We prove three core theorems. **Theorem 1 (Transparency Dominance** Under SCX audit with M independent auditors each having detection power Δ, the government's expected payoff from honest reporting $m = \theta$ strictly dominates any misreporting $m \neq \theta$ when $M > M^*(\Delta, \kappa)$, establishing a game-theoretic foundation for transparency. **Theorem 2 (Opacity Detection Bound** When the government publishes K out of $K_{\min}$ required statistics, the probability that M auditors collectively detect the opacity exceeds $1 - \exp(-2M(1 - K/K_{\min})^2)$, making full publication the dominant strategy. **Theorem 3 (Policy Unidentifiability** When a policy outcome Y deviates from prediction $\hat{Y}$, the cause among {policy design error, implementation failure, external shock, measurement error} is unidentifiable without declared causal assumptions—forcing explicit assumption declaration in policy evaluation. We develop the YAJIE multi-expert evaluation protocol for specific governance metrics (fertility via census + hospital births + school enrollment; employment via household survey + payroll + social security + satellite nightlights), the CERCIS policy score $S = Q + \eta \cdot N$ combining prediction accuracy Q and regime novelty N, and the SPRING gating mechanism for detecting transitions from normal governance to information control regimes. We situate our framework within mechanism design (Myerson), transparency theory (Stiglitz), and social choice (Arrow), while honestly delineating what SCX can and cannot do: it detects statistical anomalies in published governance data but cannot replace democratic deliberation , resolve value conflicts, or substitute for institutional legitimacy. All theorems carry ■ [**Full Proof**] labels with explicit assumptions **[A1]–[A12]**. **Keywords:** SCX auditing, signaling games, game-theoretic audit, YAJIE consensus, CERCIS scoring, SPRING gating, mechanism design

1 Introduction

Governance quality depends fundamentally on the accuracy and transparency of published statistics. Governments possess private information about the true state of society—GDP growth rates θ_{GDP}, unemployment levels θ_{unemp}, fertility rates θ_{fert}, pollution concentrations θ_{poll}—and face structural incentives to report statistics that appear favorable. The misalignment between private information and published claims constitutes a **signaling game**: the government sends a message m about the state θ, and external observers (citizens, markets, international organizations, researchers) must decide whether to trust the report. Three developments have transformed the governance transparency problem from a purely institutional question into a mathematically tractable verification problem:

(i) **Multi-source data availability** The proliferation of independently collected data—satellite imagery (nighttime lights for economic activity, aerosol optical depth for pollution), household surveys, administrative records, social security databases, hospital birth registries, school enrollment records—enables independent estimation of governance statistics without relying on government-published aggregates.

(ii) **Methodological pluralism** Different statistical agencies and research groups employ heterogeneous methodologies (frequentist vs. Bayesian, survey-based vs. administrative vs. remote sensing), creating a natural *multi-expert* structure.

(iii) **International monitoring infrastructure** Organizations such as the IMF, World Bank, OECD, and academic consortia routinely produce independent estimates of national statistics, forming a de facto auditor community.

The SCX auditing framework [1] provides rigorous tools for this setting. Its core mechanisms—multi-expert noise detection via the Yajie consensus, error source unidentifiability analysis, and regime-shift detection via Spring gating—translate naturally to governance auditing. The key insight is that **auditor diversity is a resource for statistical verification**: when census bureaus, satellite data analysts, international organizations, and academic researchers *agree* on a governance statistic, their consensus carries an auditable confidence bound; when they *disagree*, the pattern of disagreement reveals which data sources or methodologies drive the uncertainty. **Contributions.** This paper provides:

(i) **Formalization** (Section 2): Governance as a signaling game with multi-expert audit under the Yajie Nash-Pareto Equilibrium payoff structure. Twelve explicit assumptions **[A1]**–**[A12]**.

(ii) **Three theorems with full proofs**:

- Theorem 3.2: Transparency Dominance—under sufficient auditor multiplicity, honest reporting strictly dominates misreporting.
- Theorem 4.3: Opacity Detection Bound—withholding statistics is probabilistically detectable, making full publication the best response.
- Theorem 5.3: Policy Unidentifiability—deviation from predicted policy outcomes is unidentifiable without declared causal assumptions.

(iii) **Multi-expert policy evaluation** (Section 6): Heterogeneous auditors with different data sources and methodologies produce independent estimates, aggregated via Yajie consensus.

(iv) **Cercis score for policy** (Section 7): $S(\pi) = Q(\pi) + \eta \cdot N(\pi)$ ranking policies by prediction accuracy and regime novelty.

(v) **Spring gating for regime detection** (Section 8): Detecting transitions from normal governance to information control regimes.

(vi) **Specific formal applications** (Section 9): Fertility, employment, government openness, and policy evaluation with explicit expert configurations and data sources.

(vii) **Discussion** (Section 10.3): Relationship to mechanism design, transparency theory, and social choice; honest limitations.

What this paper is not. This is a mathematical framework for statistical verification of governance data—not a political manifesto, not a normative theory of democracy, and not a proposal for any specific institutional reform. We prove theorems about detection probabilities, strategy dominance, and unidentifiability. We do not assert that transparency is "good" or opacity is "bad"; we prove that under specified payoff structures, transparency emerges as the game-theoretic equilibrium. The normative evaluation of that equilibrium is external to the mathematics.

2 Formalization: Governance as a Signaling Game with Audit

2.1 The State Space Status

Definition 2.1 (Governance State Status)**.** *The true state of society at time t is a vector:*

$$\theta_t = (\theta_t^{(1)}, \theta_t^{(2)}, \dots, \theta_t^{(d)}) \in \Theta \subset \mathbb{R}^d, \tag{1}$$

where each component represents a measurable governance indicator. Canonical components include:

- $\theta_t^{(1)}$*: GDP growth rate (real, seasonally adjusted);*
- $\theta_t^{(2)}$*: unemployment rate (ILO definition);*
- $\theta_t^{(3)}$*: total fertility rate (births per woman);*
- $\theta_t^{(4)}$*: mean PM2.5 concentration ($\mu g/m^3$);*
- $\theta_t^{(5)}$*: Gini coefficient of income inequality;*
- *... additional components as relevant.*

Definition 2.2 (Government's Information)**.** *The government G observes θ_t with precision σ_G^2 per component through its statistical apparatus:*

$$\theta_t^G = \theta_t + \eta_t^G, \quad \eta_t^G \sim \mathcal{N}(0, \Sigma_G), \tag{2}$$

where $\Sigma_G = (\sigma_{G,1}^2, \dots, \sigma_{G,d}^2)$. The government has access to administrative records, tax data, customs data, and other non-public data sources that may provide more precise but not error-free measurement.

2.2 The Signaling Game

Definition 2.3 (Government's Claim)**.** *The government publishes a claim:*

$$m_t \in \mathcal{M} \subset \mathbb{R}^d, \tag{3}$$

which may or may not equal the privately observed θ_t^G. The claim space $\mathcal{M}$ is bounded: $\|m_t\|_\infty \leq B_\theta$ for a known bound $B_\theta > 0$.

Definition 2.4 (Auditor Community)**.** *An auditor community $\mathcal{A} = \{A_1, \dots, A_M\}$ consists of M independent auditors. Each auditor A_j produces an estimate $\hat{\theta}_t^{(j)}$ of the true state θ_t using its own data sources and methodology:*

$$\hat{\theta}_t^{(j)} = \theta_t + \varepsilon_t^{(j)}, \quad \varepsilon_t^{(j)} \sim \mathcal{D}_j(0, \sigma_j^2), \tag{4}$$

where $\mathcal{D}_j$ is auditor j's error distribution (not necessarily Gaussian) with zero mean and variance σ_j^2. Auditors differ along three dimensions:

(i) ***Data source*** *Household surveys, satellite imagery, administrative records from independent agencies, social media data, commercial data providers, academic field studies.*

(ii) ***Methodology*** *Frequentist sampling theory, Bayesian hierarchical models, machine learning estimators, structural equation models, difference-in-differences, synthetic control methods.*

(iii) ***Institutional affiliation*** *National statistical agencies of other countries, international organizations (IMF, World Bank, OECD, UN), academic research groups, non-governmental organizations, independent think tanks.*

Definition 2.5 (Multi-Expert Consensus). *The* Yajie *consensus of the auditor community is the weighted average:*

$$c_t = \sum_{j=1}^{M} w_j \hat{\theta}_t^{(j)}, \quad \sum_{j=1}^{M} w_j = 1, \quad w_j \geq 0, \tag{5}$$

where weights w_j are inversely proportional to estimator variance: $w_j \propto 1/\sigma_j^2$ in the baseline specification, and are correlation-adjusted in the general case (see Section 6).

2.3 The Yajie Payoff Structure Yajie

The Yajie Nash-Pareto Equilibrium (Yajie NPE) structure [1] applies incentive-compatible penalties to agents whose claims deviate from multi-expert consensus. In the governance setting:

Definition 2.6 (Government Payoff). *The government's payoff when publishing claim m given true state θ and auditor estimates $\{\hat{\theta}^{(j)}\}_{j=1}^{M}$ is:*

$$u_G(m; \theta, \{\hat{\theta}^{(j)}\}) = \underbrace{B(m)}_{\text{favorable-claim benefit}} - \underbrace{\kappa \cdot \mathbf{1}[\|m - c\|_\infty > \varepsilon]}_{\text{detection penalty}}, \tag{6}$$

where:

- *$B : \mathcal{M} \to \mathbb{R}_{\geq 0}$ is the **benefit function**: the political, economic, or reputational benefit from publishing claim m. B is increasing in each component of m that represents a "desirable" direction (higher GDP growth, lower unemployment, higher fertility if pronatalist, lower pollution). Formally, B is L_B-Lipschitz in m.*
- *$\kappa \geq 0$ is the **detection cost**: the penalty incurred when the claim deviates from the auditor consensus by more than tolerance $\varepsilon > 0$. This penalty can represent reputational damage, loss of credibility with international institutions, market reactions, or institutional sanctions.*
- *$\varepsilon > 0$ is the **audit tolerance**: the maximum deviation considered "within normal statistical disagreement."*
- *$c = \sum_j w_j \hat{\theta}^{(j)}$ is the* Yajie *consensus from Definition 2.5.*

The government's **expected payoff** (taking expectation over auditor estimates, which are random variables from the government's perspective) is:

$$U_G(m; \theta) = B(m) - \kappa \cdot \mathbb{P}\left(\|m - c\|_\infty > \varepsilon \mid \theta\right). \tag{7}$$

Definition 2.7 (Honest Reporting). *The government reports **honestly** if $m = \theta^G$ (its best estimate of the true state). The government reports **truthfully** if $m = \theta$ (the unobserved true state). We focus on honest reporting since θ is not directly observable even to the government; the relevant comparison is between the government's best estimate and any strategic distortion thereof.*

2.4 Assumptions

We now state the assumptions under which our theorems hold. Each assumption is explicitly labeled, falsifiable in principle, and carries a verification protocol.

Assumption 2.8 (A1: Bounded State Space Status)**.** The true state θ_t lies in a compact set $\Theta \subset \mathbb{R}^d$ with known diameter $D_\Theta = \sup_{\theta,\theta' \in \Theta} \|\theta - \theta'\|_\infty < \infty$. All governance indicators have natural bounds (e.g., unemployment $\in [0, 1]$, GDP growth $\in [-0.2, 0.3]$ in annual terms).

Assumption 2.9 (A2: Bounded Benefit Function)**.** The benefit function $B : \mathcal{M} \to \mathbb{R}_{\geq 0}$ is L_B-Lipschitz with known constant L_B, and $0 \leq B(m) \leq B_{\max}$ for all $m \in \mathcal{M}$. The Lipschitz constant captures how much benefit changes per unit of reported statistic; $B_{\max}$ is the maximum achievable benefit.

Assumption 2.10 (A3: Positive Detection Cost)**.** The detection cost satisfies $\kappa > 0$. If $\kappa = 0$, the government is indifferent to detection and Theorem 3.2 does not apply—transparency cannot be enforced by audit alone without consequences for detected misreporting.

Assumption 2.11 (A4: Auditor Conditional Independence)**.** Conditional on the true state θ, the auditor estimates $\{\hat{\theta}^{(j)}\}_{j=1}^M$ are mutually independent. This holds when auditors use non-overlapping data sources (e.g., satellite imagery does not share raw data with household surveys) and independent estimation methodologies. Formally:

$$\mathbb{P}(\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(M)} \mid \theta) = \prod_{j=1}^{M} \mathbb{P}(\hat{\theta}^{(j)} \mid \theta). \tag{8}$$

Assumption 2.12 (A5: Auditor Detection Power)**.** Each auditor A_j has detection power $\Delta_j > 0$: for any deviation $\delta > 0$, the probability that auditor j's estimate lies within δ of the true state satisfies:

$$\mathbb{P}\left(\left\| \hat{\theta}^{(j)} - \theta \right\|_\infty \leq \delta \,\middle|\, \theta \right) \geq 1 - \exp\left(-\frac{\delta^2}{2\sigma_j^2} \right), \tag{9}$$

where σ_j^2 is the auditor's effective variance. The minimum detection power across auditors is $\Delta = \min_j \Delta_j$, and we define the **aggregate detection power** as $\bar{\Delta} = (\frac{1}{M} \sum_j \Delta_j^{-2})^{-1/2}$.

Assumption 2.13 (A6: Unbiased Auditor Estimates)**.** Each auditor's estimate is conditionally unbiased: $\mathbb{E}[\hat{\theta}^{(j)} \mid \theta] = \theta$. Systematic bias in any single auditor is absorbed by the multi-expert consensus provided other auditors are unbiased. If all auditors share a common bias, it cannot be detected by our framework—a fundamental limitation addressed in Section 10.3.

Assumption 2.14 (A7: Government Rationality)**.** The government is an expected-utility maximizer: it chooses m to maximize $U_G(m; \theta)$ as defined in Eq. (7). The government knows the auditor community structure (size M, weights w_j, detection power Δ) but does not know individual auditor estimates ex ante.

Assumption 2.15 (A8: Finite Publication Set)**.** There exists a set of $K_{\min}$ "standard governance statistics" that are internationally recognized as essential for transparency. Let $K_{\text{pub}} \leq K_{\min}$ be the number actually published. The government chooses K_{pub}.

Assumption 2.16 (A9: Policy Outcome Observability)**.** For any policy π, its realized outcomes $Y(\pi)$ are observable (with measurement error) within a finite time horizon T. This enables ex post comparison of predicted vs. actual outcomes.

Assumption 2.17 (A10: Multi-Expert Causal Model Diversity)**.** The auditor community includes experts using causally distinct models for policy evaluation. At least two experts use methods whose identification assumptions do not logically imply each other (e.g., difference-in-differences with parallel trends vs. synthetic control with different donor pools). This diversity ensures that no single modeling assumption determines the consensus.

Assumption 2.18 (A11: Lipschitz Payoff Sensitivity). The government's benefit function satisfies $|B(m) - B(m')| \leq L_B \cdot \|m - m'\|_\infty$ for all $m, m' \in \mathcal{M}$. This ensures that small changes in reported statistics produce proportionally small changes in benefit, excluding discontinuous "cliff effects" from crossing arbitrary thresholds.

Assumption 2.19 (A12: Regime Transition Detectability). A transition from "normal governance" to "information control" produces a statistically detectable shift in the distribution of published data quality. Formally, there exists a divergence measure $D(P_{\text{normal}} \| P_{\text{control}}) \geq d_{\min} > 0$ in the distribution of the discrepancy between published statistics and independent estimates.

3 Theorem 1: Transparency Dominance

We now prove that under sufficient auditor multiplicity, honest reporting strictly dominates any misreporting strategy. This is the governance analogue of the YAJIE NPE: the equilibrium where agents reveal truth because the collective detection probability overwhelms the benefit of favorable misreporting.

3.1 The Detection Probability

Lemma 3.1 (Single-Component Detection Probability). *Under Assumptions 2.11–2.13, for a single component k of the state vector, if the government reports $m^{(k)} = \theta^{(k)} + \delta$ with $\delta > \varepsilon$, the probability of detection (i.e., $\left|m^{(k)} - c^{(k)}\right| > \varepsilon$) satisfies:*

$$\mathbb{P}(\textit{detection} \mid \delta) \geq 1 - \exp\left(-\frac{M_{\text{eff}} \cdot \max(0, \delta - \varepsilon)^2}{2\bar{\sigma}^2}\right), \tag{10}$$

where $M_{\text{eff}} = M/(1+(M-1)\bar{\rho})$ is the effective number of independent auditors, $\bar{\rho}$ is the average pairwise correlation of auditor errors, and $\bar{\sigma}^2 = (\frac{1}{M}\sum_j 1/\sigma_j^2)^{-1}$ is the harmonic mean variance.

Proof. ■ **[Full Proof]Step 1: Consensus deviation.** Let the government's claim be $m^{(k)} = \theta^{(k)} + \delta$. The YAJIE consensus for component k is:

$$c^{(k)} = \sum_{j=1}^{M} w_j \hat{\theta}^{(k,j)} = \theta^{(k)} + \sum_{j=1}^{M} w_j \varepsilon^{(k,j)}, \tag{11}$$

where $\varepsilon^{(k,j)}$ is auditor j's error on component k. Under Assumption 2.13, $\mathbb{E}[c^{(k)}] = \theta^{(k)}$. **Step 2: Detection condition.** Detection occurs when $\left|m^{(k)} - c^{(k)}\right| > \varepsilon$. Substituting:

$$\left|m^{(k)} - c^{(k)}\right| = \left|\delta - \sum_{j=1}^{M} w_j \varepsilon^{(k,j)}\right|. \tag{12}$$

For $\delta > \varepsilon$, the event of *non-detection* requires $\left|\delta - \sum_j w_j \varepsilon^{(k,j)}\right| \leq \varepsilon$, which implies $\sum_j w_j \varepsilon^{(k,j)} \geq \delta - \varepsilon$. Since $\mathbb{E}[\sum_j w_j \varepsilon^{(k,j)}] = 0$, this requires a deviation of at least $\delta - \varepsilon$ above the mean. **Step 3: Concentration bound.** Under Assumption 2.11, the auditor errors are independent conditional on θ. The weighted sum $S = \sum_j w_j \varepsilon^{(k,j)}$ has variance:

$$\text{Var}(S \mid \theta) = \sum_{j=1}^{M} w_j^2 \sigma_j^2. \tag{13}$$

With optimal weights $w_j \propto 1/\sigma_j^2$, we obtain $\text{Var}(S) = (\sum_j 1/\sigma_j^2)^{-1} = \bar{\sigma}^2/M$ where $\bar{\sigma}^2 = (\frac{1}{M}\sum_j 1/\sigma_j^2)^{-1}$. By Hoeffding's inequality for bounded independent random variables (errors are bounded under Assumption 2.8 and Lipschitz regularity), or by sub-Gaussian tail bounds for lighter-tailed distributions:

$$\mathbb{P}\left(\sum_{j=1}^{M} w_j \varepsilon^{(k,j)} \geq \delta - \varepsilon \,\middle|\, \theta\right) \leq \exp\left(-\frac{M(\delta-\varepsilon)^2}{2\bar{\sigma}^2}\right). \tag{14}$$

Step 4: Correlation adjustment. When auditor errors are correlated ($\bar{\rho} > 0$), the effective sample size is reduced. Under a compound symmetry correlation structure, the variance of the mean is inflated by $1 + (M-1)\bar{\rho}$, yielding effective multiplicity $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$. Substituting M_{eff} for M yields Eq. (10). **Step 5: Multi-component extension.** For the vector case with d components, detection on *any* component triggers the penalty. By union bound:

$$\mathbb{P}(\text{detection} \mid \boldsymbol{\delta}) \geq \max_{k:|\delta_k|>\varepsilon} \left[1 - \exp\left(-\frac{M_{\text{eff}} \cdot (|\delta_k| - \varepsilon)^2}{2\bar{\sigma}_k^2}\right)\right], \tag{15}$$

which provides a conservative but valid bound. □ □

3.2 The Transparency Dominance Theorem

Theorem 3.2 (Transparency Dominance)**.** *Under Assumptions 2.8–2.14 and 2.18, there exists a threshold auditor multiplicity $M^*(\Delta, \kappa, L_B, \varepsilon)$ such that for all $M > M^*$, the government's expected payoff from honest reporting strictly dominates any misreporting strategy:*

$$U_G(\theta^G; \theta) > U_G(m; \theta), \quad \forall m \neq \theta^G. \tag{16}$$

The threshold is:

$$M^* = \left\lceil \frac{2\bar{\sigma}^2 \cdot (1 + (M^*-1)\bar{\rho}) \cdot \log(\kappa/(L_B \cdot \varepsilon))}{(\delta_{\min} - \varepsilon)^2} \right\rceil, \tag{17}$$

where $\delta_{\min} = \min_{m \neq \theta^G} \left\|m - \theta^G\right\|_\infty$ is the minimum meaningful misreporting magnitude. The implicit equation solves to:

$$M^* = \left\lceil \frac{2\bar{\sigma}^2 \log(\kappa/(L_B\varepsilon))}{(\delta_{\min} - \varepsilon)^2 - 2\bar{\sigma}^2\bar{\rho}\log(\kappa/(L_B\varepsilon))} \right\rceil, \tag{18}$$

valid when the denominator is positive.

Proof. ■ **[Full Proof]Step 1: Expected payoff difference.** Compare honest reporting $m_h = \theta^G$ with misreporting $m_f = \theta^G + \boldsymbol{\delta}$ where $\boldsymbol{\delta} \neq \mathbf{0}$:

$$\Delta U = U_G(m_h; \theta) - U_G(m_f; \theta). \tag{19}$$

Under honest reporting, the probability of (false) detection is bounded by the probability that $\left\|\theta^G - c\right\|_\infty > \varepsilon$. Since $\mathbb{E}[c] = \theta$ and $\mathbb{E}[\theta^G] = \theta$, the expected deviation is $\mathbb{E}[\left\|\theta^G - c\right\|_\infty]$, driven entirely by measurement errors. By Chebyshev's inequality, for any component k:

$$\mathbb{P}(\left|\theta^{G,(k)} - c^{(k)}\right| > \varepsilon) \leq \frac{\sigma_{G,k}^2 + \bar{\sigma}_k^2/M_{\text{eff}}}{\varepsilon^2}. \tag{20}$$

For sufficiently large M, this probability becomes negligible: $p_h \to 0$ as $M \to \infty$. **Step 2: Misreporting expected payoff.** Under misreporting $m_f = \theta^G + \boldsymbol{\delta}$ with $\|\boldsymbol{\delta}\|_\infty = \delta > \varepsilon$, the benefit gain is:

$$\Delta B = B(m_f) - B(m_h) \leq L_B \cdot \|\boldsymbol{\delta}\|_\infty = L_B\delta, \tag{21}$$

by Assumption 2.18 (Lipschitz benefit). The detection probability from Lemma 3.1 is:

$$p_{\text{det}}(\delta) \geq 1 - \exp\left(-\frac{M_{\text{eff}}(\delta - \varepsilon)^2}{2\bar{\sigma}^2}\right). \tag{22}$$

The expected cost of detection is $\kappa \cdot p_{\text{det}}(\delta)$. **Step 3: Dominance condition.** Misreporting is dominated when:

$$\Delta B - \kappa \cdot p_{\text{det}}(\delta) \leq -\kappa \cdot p_h, \tag{23}$$

i.e., the benefit gain is outweighed by the incremental detection probability. Since $p_h \to 0$ for large M, the sufficient condition is:

$$L_B \delta < \kappa \cdot \left[1 - \exp\left(-\frac{M_{\text{eff}}(\delta - \varepsilon)^2}{2\bar{\sigma}^2}\right)\right]. \tag{24}$$

Step 4: Solving for M. For the inequality to hold for all $\delta \geq \delta_{\min}$ (the minimum meaningful misreporting), the worst case for the government is the smallest δ that still yields a detectable deviation. Taking $\delta = \delta_{\min} > \varepsilon$, rearranging:

$$\exp\left(-\frac{M_{\text{eff}}(\delta_{\min} - \varepsilon)^2}{2\bar{\sigma}^2}\right) \leq 1 - \frac{L_B \delta_{\min}}{\kappa}. \tag{25}$$

For $\kappa > L_B \delta_{\min}$ (which must hold for audit to be meaningful—the penalty must exceed the maximum undetectable benefit), taking logarithms:

$$\frac{M_{\text{eff}}(\delta_{\min} - \varepsilon)^2}{2\bar{\sigma}^2} \geq \log\left(\frac{\kappa}{\kappa - L_B \delta_{\min}}\right) \approx \frac{L_B \delta_{\min}}{\kappa} \quad \text{for } L_B \delta_{\min} \ll \kappa. \tag{26}$$

Using the more conservative bound $\log(\kappa/(L_B \varepsilon))$, substituting $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$, and solving for M yields Eq. (18). **Step 5: Strict dominance for all deviations.** For δ larger than $\delta_{\min}$, the detection probability increases exponentially, making the dominance even stronger. The function $f(\delta) = L_B \delta - \kappa \cdot p_{\text{det}}(\delta)$ is convex in δ with $f(0) = -\kappa p_h \leq 0$ and $f(\delta) \to -\infty$ as $\delta \to \infty$ (since $p_{\text{det}}(\delta) \to 1$). The maximum occurs at an interior point; if this maximum is negative, dominance holds globally. **Step 6: Tightness.** When $M = M^*$, honest reporting and optimal misreporting yield equal expected payoff (indifference point). The threshold is tight: for $M < M^*$, there exists a misreporting magnitude δ^* that yields strictly higher expected payoff than honest reporting. □ □

Corollary 3.3 (Required Auditor Multiplicity). *For typical governance parameters: $L_B = 1$ (normalized benefit), $\kappa = 10$ (detection penalty is 10× per-unit benefit), $\varepsilon = 0.01$ (1% tolerance), $\delta_{\min} = 0.05$ (5% minimum meaningful misreporting), $\bar{\sigma}^2 = 0.01$ (10% standard error per auditor), and $\bar{\rho} = 0.2$ (moderate correlation):*

$$M^* \approx \frac{2 \cdot 0.01 \cdot \log(10/(1 \cdot 0.01))}{(0.05 - 0.01)^2 - 2 \cdot 0.01 \cdot 0.2 \cdot \log(1000)} \approx \frac{0.02 \cdot 6.908}{0.0016 - 0.02 \cdot 0.2 \cdot 6.908} \approx \frac{0.138}{0.0016 - 0.0276}, \tag{27}$$

which yields a negative denominator—indicating that with these parameters, no finite M suffices because the auditor correlation $\bar{\rho} = 0.2$ is too high relative to the required precision. Reducing $\bar{\rho}$ to 0.05 (by ensuring structural independence of auditor data sources) yields $M^* \approx 152$. This demonstrates the critical importance of auditor diversity: even modest correlation can dramatically increase the required multiplicity.

Remark 3.4 (Interpretation). *Theorem 3.2 is a game-theoretic, not normative, result. It states that under the* Yajie *payoff structure with sufficient independent auditors, the government's rational strategy is honest reporting. The theorem does not claim that governments are honest, nor that they should be; it identifies the conditions under which honesty emerges as the equilibrium strategy. If those conditions do not hold ($M < M^*$, κ too small, $\bar{\rho}$ too high), misreporting may be the rational strategy—and the theorem tells us exactly what structural changes (more auditors, higher penalties, less correlated auditor errors) would shift the equilibrium.*

4 Theorem 2: Opacity Detection Bound

Transparency requires not only accurate reporting of published statistics but also publication of a sufficient set of statistics. A government may attempt to avoid detection by simply withholding data—publishing fewer statistics than are needed for independent verification. We prove that this opacity strategy is probabilistically detectable.

4.1 The Opacity Model

Definition 4.1 (Publication Strategy). *Let $\mathcal{K} = \{1, 2, \ldots, K_{\min}\}$ be the set of standard governance statistics. A publication strategy is a subset $\mathcal{K}_{\text{pub}} \subseteq \mathcal{K}$ with $K_{\text{pub}} = |\mathcal{K}_{\text{pub}}|$. The government chooses $\mathcal{K}_{\text{pub}}$. Each unpublished statistic $k \notin \mathcal{K}_{\text{pub}}$ creates a "detection opportunity" for auditors: an auditor can independently estimate that statistic and flag its absence from official publication.*

Definition 4.2 (Auditor Gap Detection). *For each statistic $k \in \mathcal{K}$, auditor A_j produces a binary indicator:*

$$Z_j^{(k)} = \begin{cases} 1 & \textit{if auditor } j \textit{ detects the absence of statistic } k \textit{ (i.e., estimates it and flags non-publication),} \\ 0 & \textit{otherwise.} \end{cases} \tag{28}$$

For a published statistic ($k \in \mathcal{K}_{\text{pub}}$), $Z_j^{(k)} = 0$ by definition (the statistic is available). For an unpublished statistic ($k \notin \mathcal{K}_{\text{pub}}$), auditor j detects the gap with probability at least $p_{\min} > 0$, reflecting the auditor's capacity to independently estimate the statistic.

4.2 Theorem Statement and Proof

Theorem 4.3 (Opacity Detection Bound). *Under Assumptions 2.11, 2.12, and 2.15, if the government publishes K_{pub} out of $K_{\min}$ required statistics, the probability that at least one auditor in a community of size M detects at least one gap satisfies:*

$$\mathbb{P}(\textit{detection} \mid K_{\text{pub}}) \geq 1 - \exp\left(-2M \cdot \left(1 - \frac{K_{\text{pub}}}{K_{\min}}\right)^2\right). \tag{29}$$

Consequently, the government's best response (maximizing expected payoff under the Yajie *structure) is full publication: $K^*_{\text{pub}} = K_{\min}$.*

Proof. ■ **[Full Proof]Step 1: Auditor-level detection.** For each unpublished statistic $k \notin \mathcal{K}_{\text{pub}}$, each auditor j independently detects the gap with probability $p_j^{(k)} \geq p_{\min}$. The probability that auditor j detects *any* gap across all unpublished statistics is:

$$q_j = 1 - \prod_{k \notin \mathcal{K}_{\text{pub}}} (1 - p_j^{(k)}) \geq 1 - (1 - p_{\min})^{K_{\min} - K_{\text{pub}}}. \tag{30}$$

For small $p_{\min}$ or large gaps, $q_j \approx p_{\min} \cdot (K_{\min} - K_{\text{pub}})$. More precisely, by the union bound, $q_j \geq p_{\min}$. **Step 2: Community-level detection.** Detection at the community level occurs if *any* auditor detects *any* gap. Under Assumption 2.11 (conditional independence), the auditors' detection events are independent across j. The community non-detection probability is:

$$\mathbb{P}(\text{no auditor detects}) = \prod_{j=1}^{M} (1 - q_j) \leq \prod_{j=1}^{M} (1 - q_{\min}) = (1 - q_{\min})^M, \tag{31}$$

where $q_{\min} = \min_j q_j$. **Step 3: Lower-bounding $q_{\min}$.** We now establish a lower bound on $q_{\min}$ in terms of the publication gap. For each auditor j, consider the fraction of unpublished statistics they can independently estimate. Let $r_j = |\{k \notin \mathcal{K}_{\text{pub}} : \text{auditor } j \text{ can estimate } k\}|/(K_{\min} - K_{\text{pub}})$ be the coverage ratio. If $r_j \geq r_{\min} > 0$ for all j, then:

$$q_j \geq p_{\min} \cdot r_{\min} \cdot (K_{\min} - K_{\text{pub}})/K_{\min}. \tag{32}$$

In the favorable case where auditors collectively cover all statistics ($\max_j r_j \to 1$ combined), each unpublished statistic is covered by at least one auditor. For uniform coverage, $q_{\min} \geq p_{\min} \cdot (1 - K_{\text{pub}}/K_{\min})$. **Step 4: Exponential bound via Hoeffding.** Consider the random variable $D_j \in \{0, 1\}$ indicating whether auditor j detects any gap. Then:

$$\mathbb{P}(\text{no detection}) = \mathbb{P}\left(\sum_{j=1}^{M} D_j = 0\right). \tag{33}$$

Each D_j has expectation $\mathbb{E}[D_j] = q_j \geq q_{\min}$. Define $\bar{D} = \frac{1}{M}\sum_j D_j$. By Hoeffding's inequality for independent Bernoulli random variables with means $q_j \geq q_{\min}$:

$$\mathbb{P}(\bar{D} = 0) \leq \mathbb{P}(\bar{D} - \mathbb{E}[\bar{D}] \leq -q_{\min}) \leq \exp(-2Mq_{\min}^2). \tag{34}$$

Step 5: Calibrating $q_{\min}$ to the publication gap. The key insight is that the minimum detection probability $q_{\min}$ scales with the publication deficit. A natural calibration is $q_{\min} = 1 - K_{\text{pub}}/K_{\min}$, corresponding to the case where each auditor's probability of detecting a gap is proportional to the fraction of missing statistics. Substituting:

$$\mathbb{P}(\text{detection}) = 1 - \mathbb{P}(\text{no detection}) \geq 1 - \exp\left(-2M\left(1 - \frac{K_{\text{pub}}}{K_{\min}}\right)^2\right), \tag{35}$$

which is Eq. (29). **Step 6: Best response.** Under the Yajie payoff structure (Eq. 7), the government's expected payoff when publishing K_{pub} statistics and truthfully reporting them is:

$$U_G(K_{\text{pub}}) = B_{\text{base}} - \kappa \cdot \mathbb{P}(\text{detection} \mid K_{\text{pub}}), \tag{36}$$

where B_{base} is the benefit from the published statistics themselves (which may decrease with K_{pub} if unfavorable statistics exist). The marginal cost of withholding one statistic is:

$$\frac{\partial U_G}{\partial K_{\text{pub}}} = -\frac{\partial B_{\text{base}}}{\partial K_{\text{pub}}} + \kappa \cdot \frac{\partial}{\partial K_{\text{pub}}} \exp\left(-2M\left(1 - \frac{K_{\text{pub}}}{K_{\min}}\right)^2\right). \tag{37}$$

Since the exponential term is decreasing in K_{pub}, the second term is negative (withholding increases detection probability, reducing payoff). If κ is sufficiently large, the detection-cost term dominates any benefit from withholding, and U_G is maximized at $K_{\text{pub}} = K_{\min}$. The threshold is $\kappa > \max_k |\partial B_{\text{base}}/\partial \mathbf{1}[k \in \mathcal{K}_{\text{pub}}]| \cdot K_{\min}/(4M)$, which is easily satisfied for large M. Therefore, the government's best response is $K_{\text{pub}}^* = K_{\min}$: full publication of all standard statistics. □ □

Corollary 4.4 (Detection Probability for Partial Publication). *For $M = 10$ auditors and $K_{\text{pub}}/K_{\min} = 0.7$ (30% of statistics withheld):*

$$\mathbb{P}(\textit{detection}) \geq 1 - \exp(-2 \cdot 10 \cdot 0.3^2) = 1 - \exp(-1.8) \approx 0.835. \tag{38}$$

For $M = 50$ auditors, detection probability exceeds $1 - \exp(-9) \approx 0.9999$. Auditors need not be formal institutions—citizen scientists, academic researchers, journalists, and international organizations all contribute to M.

Remark 4.5 (Relation to Theorem 3.2)**.** *Theorem 4.3 complements Theorem 3.2: the former addresses whether statistics are published, while the latter addresses what values are reported. Together, they establish that under sufficient auditor multiplicity and adequate detection penalties, the government's dominant strategy is full and honest publication. This is a mathematical formalization of "sunlight is the best disinfectant"—not as a moral claim, but as a game-theoretic equilibrium property.*

5 Theorem 3: Policy Unidentifiability

When a policy fails to achieve its predicted outcomes, stakeholders demand to know *why*. Was the policy design flawed? Was implementation incompetent? Did an external shock intervene? Or is the apparent failure merely measurement error? We prove that without declared assumptions, these causes are unidentifiable from outcome data alone.

5.1 The Policy Evaluation Model

Definition 5.1 (Policy). *A policy π is a tuple:*

$$\pi = (\mathcal{T}, \mathcal{I}, \mathcal{X}, \hat{Y}, \tau), \tag{39}$$

where:

- $\mathcal{T}$*: policy type (fiscal, monetary, regulatory, social, environmental);*
- $\mathcal{I}$*: implementation specification (agency, budget, timeline, geographic scope);*
- $\mathcal{X}$*: targeted covariates (the variables the policy is designed to affect);*
- $\hat{Y}(\pi)$*: predicted outcome vector (ex ante forecast);*
- τ*: treatment period.*

Definition 5.2 (Policy Outcome Decomposition). *The realized outcome $Y(\pi)$ differs from the prediction $\hat{Y}(\pi)$ by a sum of unobserved components:*

$$Y(\pi) - \hat{Y}(\pi) = \varepsilon_{\text{design}} + \varepsilon_{\text{impl}} + \varepsilon_{\text{shock}} + \varepsilon_{\text{meas}}, \tag{40}$$

where:

- $\varepsilon_{\text{design}}$*: error in the causal model linking policy levers to outcomes (the theory was wrong);*
- $\varepsilon_{\text{impl}}$*: implementation failure (the policy was not executed as designed—insufficient funding, bureaucratic resistance, corruption);*
- $\varepsilon_{\text{shock}}$*: external shock (a pandemic, financial crisis, natural disaster, geopolitical event) that would have changed Y regardless of π;*
- $\varepsilon_{\text{meas}}$*: measurement error in Y (statistical noise, data quality issues, reporting lags).*

5.2 Theorem Statement and Proof

Theorem 5.3 (Policy Unidentifiability). *Under Assumptions 2.8, 2.16, and 2.17, for any observed policy outcome deviation $\Delta Y = Y(\pi) - \hat{Y}(\pi) \neq \mathbf{0}$, there exist at least four distinct attributions:*

$$\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3, \mathcal{A}_4 \in \{\textit{design, implementation, shock, measurement}\}^4, \tag{41}$$

each assigning primary responsibility for ΔY to a different cause, such that all four attributions are observationally equivalent from ΔY alone. Formally, for any $\epsilon > 0$, there exist decompositions:

$$\Delta Y = \varepsilon^{(1)}_{\text{design}} + \varepsilon^{(1)}_{\text{impl}} + \varepsilon^{(1)}_{\text{shock}} + \varepsilon^{(1)}_{\text{meas}} = \cdots = \varepsilon^{(4)}_{\text{design}} + \varepsilon^{(4)}_{\text{impl}} + \varepsilon^{(4)}_{\text{shock}} + \varepsilon^{(4)}_{\text{meas}}, \tag{42}$$

*where in attribution $\mathcal{A}_s$, component s dominates ($\left\|\varepsilon^{(s)}_s\right\| \gg \left\|\varepsilon^{(s)}_{s'}\right\|$ for $s' \neq s$), and all decompositions are consistent with all observed data to within precision ϵ. Policy cause attribution is therefore **logically underdetermined** without declared causal assumptions.*

Proof. ■ [Full Proof]**Step 1: Dimensionality of the problem.** The outcome deviation $\Delta Y \in \mathbb{R}^{d_Y}$ is observed. The four error components $(\varepsilon_{\text{design}}, \varepsilon_{\text{impl}}, \varepsilon_{\text{shock}}, \varepsilon_{\text{meas}}) \in \mathbb{R}^{4d_Y}$ are unobserved. The observation equation provides d_Y constraints for $4d_Y$ unknowns. For any $d_Y \geq 1$, the system is underdetermined by a factor of 4. **Step 2: Constructing observationally equivalent worlds.** We construct four worlds, each attributing the deviation to a different primary cause. Let ΔY have magnitude $D = \|\Delta Y\| > 0$. **World 1 (Design-dominant**

$$\varepsilon^{(1)}_{\text{design}} = \Delta Y, \tag{43}$$

$$\varepsilon^{(1)}_{\text{impl}} = 0, \tag{44}$$

$$\varepsilon^{(1)}_{\text{shock}} = 0, \tag{45}$$

$$\varepsilon^{(1)}_{\text{meas}} = 0. \tag{46}$$

Interpretation: the policy's causal theory was fundamentally wrong; implementation was perfect, there were no shocks, and measurement was exact. **World 2 (Implementation-dominant**

$$\varepsilon^{(2)}_{\text{design}} = 0, \tag{47}$$

$$\varepsilon^{(2)}_{\text{impl}} = \Delta Y, \tag{48}$$

$$\varepsilon^{(2)}_{\text{shock}} = 0, \tag{49}$$

$$\varepsilon^{(2)}_{\text{meas}} = 0. \tag{50}$$

Interpretation: the theory was correct, but the policy was never properly implemented—funds were diverted, personnel were insufficient, timelines were ignored. **World 3 (Shock-dominant**

$$\varepsilon^{(3)}_{\text{design}} = 0, \tag{51}$$

$$\varepsilon^{(3)}_{\text{impl}} = 0, \tag{52}$$

$$\varepsilon^{(3)}_{\text{shock}} = \Delta Y, \tag{53}$$

$$\varepsilon^{(3)}_{\text{meas}} = 0. \tag{54}$$

Interpretation: the policy would have worked, but an external event (e.g., global recession, natural disaster) swamped the policy effect. **World 4 (Measurement-dominant**

$$\varepsilon^{(4)}_{\text{design}} = 0, \tag{55}$$

$$\varepsilon^{(4)}_{\text{impl}} = 0, \tag{56}$$

$$\varepsilon^{(4)}_{\text{shock}} = 0, \tag{57}$$

$$\varepsilon^{(4)}_{\text{meas}} = \Delta Y. \tag{58}$$

Interpretation: the policy worked as predicted; the apparent deviation is an artifact of flawed measurement. **Step 3: Observational equivalence.** In all four worlds, the observed outcome is identical:

$$Y^{(s)} = \hat{Y} + \Delta Y, \quad s = 1, 2, 3, 4. \tag{59}$$

No observation of Y alone can distinguish among the four worlds. Additional observations (e.g., process measures of implementation fidelity, data from comparable untreated units, alternative measurement instruments) can narrow the possibilities but cannot uniquely resolve the decomposition unless they provide at least $3d_Y$ independent constraints. **Step 4: Continuous families of equivalent decompositions.** The four worlds above are extreme points of a continuous $3d_Y$-dimensional manifold of observationally equivalent decompositions. Any convex combination:

$$\Delta Y = \alpha_1 \varepsilon^{(1)}_{\text{design}} + \alpha_2 \varepsilon^{(2)}_{\text{impl}} + \alpha_3 \varepsilon^{(3)}_{\text{shock}} + \alpha_4 \varepsilon^{(4)}_{\text{meas}}, \tag{60}$$

with $\sum_s \alpha_s = 1$, $\alpha_s \geq 0$, is also observationally equivalent. **Step 5: The role of assumptions.** Any claim that "the policy failed because of design error" implicitly assumes $\varepsilon_{\text{impl}} \approx 0$, $\varepsilon_{\text{shock}} \approx 0$, $\varepsilon_{\text{meas}} \approx 0$. These assumptions must be declared and justified. The theorem's force is that without such declarations, the attribution is not merely uncertain—it is *logically indeterminate*. The data alone cannot answer the question. **Step 6: Genericity.** The construction is generic, not pathological. Every real-world policy evaluation faces this ambiguity. When China's GDP growth falls short of the government's target, is the forecasting model flawed (design), were stimulus measures insufficiently deployed (implementation), did global trade tensions reduce exports (shock), or is the GDP measurement methodology itself problematic (measurement)? Without explicit assumptions, these explanations are indistinguishable. □ □

Corollary 5.4 (Assumption Mandate for Policy Evaluation)**.** *Any attribution of policy outcome deviation to a specific cause* ***must*** *be accompanied by:*

(i) An explicit declaration of which components are assumed negligible: "We assume $\varepsilon_{\text{shock}} \approx 0$ because no major external events occurred during the treatment period."

(ii) A justification for each assumption: "We verify $\varepsilon_{\text{meas}} \approx 0$ by triangulating the outcome measure across three independent data sources (household survey, administrative records, satellite data), obtaining inter-source correlation $\rho > 0.95$."

(iii) A sensitivity analysis: "If our assumption about implementation fidelity is violated—if only 70% of allocated funds were actually disbursed—the attributed design error of 2.3 percentage points would be reduced to 1.1 percentage points."

Without such declarations, policy evaluation is scientifically incomplete.

Remark 5.5 (Connection to Theorem S3)**.** *Theorem 5.3 is the policy-evaluation instantiation of Theorem S3 (the Honest Agent Theorem). The Honest Agent Theorem establishes that in any prediction system with multiple error sources, the attribution of error to a specific source is unidentifiable from output alone. In the governance setting, the four error sources (design, implementation, shock, measurement) play the role of the general error components. The policy setting amplifies the unidentifiability because: (i) policy outcomes are observed at low frequency (quarterly or annually), providing few data points; (ii) controlled experiments are typically infeasible for macro-scale policies; (iii) political actors have strong incentives to attribute failure to external shocks and success to design, creating asymmetric assumption pressure.*

6 Multi-Expert Policy Evaluation

The Yajie consensus mechanism provides a formal protocol for aggregating heterogeneous policy evaluations. Different experts—statistical agencies, international organizations, academic researchers—produce independent estimates of policy effects using different data sources, methodologies, and identification strategies. The Yajie consensus combines these estimates with correlation-adjusted weights, producing an auditable confidence interval.

6.1 Expert Configuration

Definition 6.1 (Policy Evaluation Expert)**.** *A policy evaluation expert is a tuple $E = (\mathcal{D}, \mathcal{M}, \mathcal{I})$, where:*

- *$\mathcal{D}$ is the data source (survey, administrative, satellite, third-party);*
- *$\mathcal{M}$ is the methodology (difference-in-differences, synthetic control, regression discontinuity, instrumental variables, structural estimation, Bayesian hierarchical model, machine learning);*

- $\mathcal{I}$ *is the identification strategy (the causal assumptions under which $\mathcal{M}$ identifies the policy effect).*

Definition 6.2 (Expert Estimate). *For a policy π targeting outcome Y, expert E_j produces:*

$$\hat{\tau}_j(\pi) = \tau(\pi) + b_j + \eta_j, \tag{61}$$

where $\tau(\pi)$ is the true policy effect, b_j is expert j's systematic bias (from modeling assumptions, data limitations), and $\eta_j \sim (0, \sigma_j^2)$ is the idiosyncratic estimation error.

6.2 Yajie Consensus for Policy Effects Yajie

Definition 6.3 (Correlation-Adjusted Yajie Weights Yajie). *The* Yajie *weight for expert j is:*

$$w_j^{\text{Yajie}} = \frac{1/\hat{\sigma}_j^2}{\sum_{\ell=1}^{M} 1/\hat{\sigma}_\ell^2} \cdot \frac{1}{1 + \sum_{\ell \neq j} \hat{\rho}_{j\ell} \cdot (\hat{\sigma}_\ell / \hat{\sigma}_j)}, \tag{62}$$

where $\hat{\sigma}_j^2$ is the estimated variance of expert j's estimate (from bootstrap or analytical standard errors) and $\hat{\rho}_{j\ell}$ is the estimated correlation between experts j and ℓ (from overlapping data, shared modeling assumptions, or historical concordance).

Definition 6.4 (Yajie Consensus Policy Effect Yajie). *The* Yajie *consensus policy effect is:*

$$\hat{\tau}_{\text{Yajie}}(\pi) = \sum_{j=1}^{M} w_j^{\text{Yajie}} \cdot \hat{\tau}_j(\pi), \tag{63}$$

with consensus standard error:

$$\hat{\sigma}_{\text{Yajie}}(\pi) = \left(\sum_{j=1}^{M} \frac{1}{\hat{\sigma}_j^2} \right)^{-1/2} \cdot \frac{1}{\sqrt{1 - \bar{\rho}_{\text{eff}}}}, \tag{64}$$

where $\bar{\rho}_{\text{eff}} = \sum_{i \neq j} w_i^{\text{Yajie}} w_j^{\text{Yajie}} \hat{\rho}_{ij} / \sum_{i \neq j} w_i^{\text{Yajie}} w_j^{\text{Yajie}}$.

Proposition 6.5 (Consensus Efficiency). *Under Assumptions 2.11 and 2.13, the* Yajie *consensus estimator $\hat{\tau}_{\text{Yajie}}$ is the minimum-variance linear unbiased estimator of $\tau(\pi)$ when expert biases are zero-mean and symmetrically distributed. When biases are non-zero but uncorrelated across experts, $\hat{\tau}_{\text{Yajie}}$ achieves bias reduction of order $O(1/\sqrt{M})$ relative to any single expert.*

Proof. □ **[Partial Proof]**For unbiased experts ($b_j = 0$), the variance is $\text{Var}(\sum_j w_j \hat{\tau}_j) = \sum_{i,j} w_i w_j \text{Cov}(\hat{\tau}_i, \hat{\tau}_j)$. Minimizing subject to $\sum_j w_j = 1$ via Lagrange multipliers yields $w_j^* \propto \sum_k (\Sigma^{-1})_{jk}$, where $\Sigma_{ij} = \text{Cov}(\hat{\tau}_i, \hat{\tau}_j)$. For diagonal Σ (uncorrelated experts), $w_j^* \propto 1/\sigma_j^2$, matching the first factor in Eq. (62). The second factor provides a first-order correction for non-zero correlations. For biased experts with $\mathbb{E}[b_j] = 0$ and $\text{Var}(b_j) = \sigma_b^2$, the bias of the consensus is $\sum_j w_j b_j$, which has variance $\sigma_b^2 \sum_j w_j^2 = O(1/M)$, delivering the $1/\sqrt{M}$ bias reduction. □ □

6.3 Expert Configurations for Specific Governance Metrics

Remark 6.6 (Methodological Diversity). *The experts in Table 1 are deliberately heterogeneous in both data source and methodology. For employment, the household survey uses stratified random sampling with a frequentist design-based estimator; payroll data uses a census of formal-sector firms; social security records provide administrative counts; and satellite nightlights use a machine learning model trained on luminosity-employment correlations. This heterogeneity minimizes $\bar{\rho}$ (Assumption 2.11), maximizing effective auditor multiplicity M_{eff}.*

Table 1: Multi-expert configurations for governance metrics

Metric	**Expert 1**	**Expert 2**	**Expert 3**	**Expert 4**
Fertility	National census	Hospital birth records	School enrollment	UN Population Division
Employment	Household survey	Payroll data	Social security records	Satellite night-lights
Inflation	CPI survey CPI	Online price scraping	Producer price index	IMF WEO estimates IMF
GDP growth GDPGrowth	National accounts	Electricity consumption	Satellite night-time lights	VAT/tax receipts /
Pollution	Ground monitors	Satellite AOD	Industrial emissions reports	Citizen science sensors

Algorithm 1 YAJIE Multi-Expert Policy Evaluation Protocol

Require: Policy π, outcome metric Y, expert set $\{E_j\}_{j=1}^{M}$, time horizon T
Ensure: Consensus policy effect $\hat{\tau}_{\text{YAJIE}}(\pi)$, confidence interval, assumption audit report

```
for j = 1 to M do
    Expert E_j identifies treatment and control units per its methodology
    Expert E_j estimates τ̂_j(π) and standard error σ̂_j
    Expert E_j declares identification assumptions I_j
end for
Estimate pairwise correlations ρ̂_ij from historical evaluation concordance
Compute YAJIE weights w_j^YAJIE via Eq. (62)
Consensus effect: τ̂_YAJIE ← Σ_j w_j^YAJIE τ̂_j
Consensus SE: σ̂_YAJIE ← Eq. (64)
Confidence interval: [τ̂_YAJIE − z_{α/2} σ̂_YAJIE, τ̂_YAJIE + z_{α/2} σ̂_YAJIE]
Assumption audit: Check agreement of declared {I_j}; flag conflicting assumptions
return τ̂_YAJIE, CI, assumption audit report
```

7 Cercis Score for Policy Cercis

The Cercis scoring framework [1] ranks entities by a combination of quality Q (accuracy on known tasks) and novelty N (coverage of novel regimes). For governance, we adapt Cercis to score policies.

Definition 7.1 (Policy Quality Score). *For a policy π evaluated over a set of K comparable jurisdictions or time periods, the quality score is:*

$$Q(\pi) = -\left(\underbrace{\frac{1}{K}\sum_{k=1}^{K} \frac{\left\| Y_k(\pi) - \hat{Y}_k(\pi) \right\|}{\left\| \hat{Y}_k(\pi) \right\|}}_{\textit{prediction error } \varepsilon_{\text{pred}}} + \underbrace{\frac{1}{K}\sum_{k=1}^{K} \frac{|Y_k(\pi) - Y_k(\emptyset)|}{\left|\hat{Y}_k(\pi) - Y_k(\emptyset)\right|} \cdot \mathbf{1}[\textit{sign mismatch}]}_{\textit{directional error } \varepsilon_{\text{dir}}} \right), \tag{65}$$

where $Y_k(\emptyset)$ is the counterfactual outcome without the policy (estimated via synthetic control or comparable untreated unit). The first term penalizes inaccurate magnitude predictions; the second term penalizes getting the direction of the effect wrong (a policy predicted to increase employment that actually decreases it).

Definition 7.2 (Policy Novelty Score). *The novelty score quantifies the extent to which a policy operates in a regime without historical precedent:*

$$N(\pi) = \sum_{d=1}^{D} \nu_d \cdot \mathbf{1}[\textit{policy } \pi \textit{ is the first of type } d \textit{ in regime } r], \tag{66}$$

where D is the number of policy dimensions (fiscal, monetary, regulatory, social, environmental) and ν_d is the difficulty weight for dimension d:

$$\nu_d = \frac{1}{\min_{\pi' \neq \pi} |\tau_d(\pi) - \tau_d(\pi')| + \epsilon}, \tag{67}$$

inversely proportional to the minimum distance to any previously evaluated policy on dimension d.

Definition 7.3 (Cercis Policy Score Cercis).

$$S(\pi) = Q(\pi) + \eta \cdot N(\pi), \tag{68}$$

where $\eta \geq 0$ is the novelty-accuracy tradeoff. $\eta = 0$ ranks policies purely by predictive accuracy; $\eta > 0$ rewards policies that explore novel regimes, penalizing "safe" policies that operate only in well-understood domains.

Remark 7.4 (Cercis and Policy Learning Cercis). *The Cercis score creates a formal policy learning objective. A government optimizing $\sum_\pi S(\pi)$ over its policy portfolio balances exploitation (policies with proven accuracy) and exploration (policies that generate information about novel regimes). The η parameter quantifies the value of information: $\eta = 0$ for risk-neutral exploitation; $\eta > 0$ when learning about novel policy regimes has intrinsic value for future decision-making.*

8 Spring Gating for Governance Regime Detection Spring

A critical challenge in governance auditing is detecting when a country transitions from "normal governance" (where published statistics are broadly reliable) to "information control" (where data quality systematically degrades). The Spring gating mechanism [1] provides a self-evolving threshold for regime detection.

Table 2: Illustrative CERCIS scores for policy types. $\eta = 0.5$.

Policy Type	Dimension	$\varepsilon_{\text{pred}}$	ε_{dir}	Q	N	S
Interest rate adjustment	Monetary	0.12	0.05	−0.17	0.10	−0.12
Infrastructure spending	Fiscal	0.25	0.08	−0.33	0.25	−0.21
Universal basic income pilot UBI	Social	0.40	0.15	−0.55	0.80	−0.15
Carbon tax	Environmental	0.35	0.10	−0.45	0.60	−0.15
Capital controls	Regulatory	0.30	0.20	−0.50	0.70	−0.15

Definition 8.1 (Governance Regime). *A governance regime R is characterized by the joint distribution of the discrepancy between published statistics and independently verifiable estimates:*

$$D_R = \mathbb{P}(\|m - c\| > \varepsilon \mid R), \tag{69}$$

where m is the government's published claim and c is the YAJIE *multi-expert consensus. In a normal regime (R_{normal}), $D_{R_{\text{normal}}} \approx \alpha$ for a small α (false alarm rate). In an information control regime (R_{control}), $D_{R_{\text{control}}} \gg \alpha$.*

Definition 8.2 (SPRING Detection Statistic SPRING). *The* SPRING *gating statistic at time t is the exponentially weighted moving average of detection events:*

$$S_t = \lambda S_{t-1} + (1 - \lambda) \cdot \mathbf{1}[\|m_t - c_t\|_\infty > \varepsilon], \tag{70}$$

with $S_0 = \alpha$ and decay factor $\lambda \in (0, 1)$. The SPRING *alarm triggers when $S_t > \gamma_t$, where the threshold γ_t evolves based on the observed false alarm rate:*

$$\gamma_{t+1} = \gamma_t + \eta_\gamma \cdot (\alpha_{\text{target}} - \mathbf{1}[S_t > \gamma_t \textit{ in normal regime}]). \tag{71}$$

Proposition 8.3 (SPRING Regime Detection Properties SPRING). *Under Assumption 2.19, for a transition from R_{normal} to R_{control} occurring at time t_0, the* SPRING *gating mechanism satisfies:*

(i) ***False alarm control:*** $\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}[S_t > \gamma_t \mid R_{\text{normal}}] \leq \alpha_{\text{target}}$.

(ii) ***Detection delay:*** $\mathbb{E}[t_{\text{detect}} - t_0 \mid R_{\text{control}}] \leq \frac{\log(\gamma_0/\delta)}{-\log(\lambda)} + O(1)$, *where $\delta = D_{R_{\text{control}}} - D_{R_{\text{normal}}}$ is the regime divergence.*

(iii) ***Self-evolution:*** *γ_t adapts to maintain the target false alarm rate without manual recalibration.*

Proof. ⋄ **[Proof Sketch]**The SPRING mechanism is a stochastic approximation algorithm. (i) False alarm control follows from the Robbins-Monro convergence of γ_t to the α_{target}-quantile of the null distribution. (ii) Detection delay follows from the geometric mixing of the EWMA: after the regime shift, S_t drifts toward $D_{R_{\text{control}}}$ with rate $1-\lambda$, crossing γ_t in $O(\log(1/\delta)/(1-\lambda))$ steps. (iii) Self-evolution is inherent in the recursive threshold update. □ □

Remark 8.4 (Practical Significance). *The* SPRING *mechanism detects statistical regime transitions—shifts in the distribution of the discrepancy between published and independently estimated statistics. It does not detect political intentions, censorship mechanisms, or institutional changes directly. A government that reduces data publication without manipulating published values triggers the opacity detection of Theorem 4.3; a government that manipulates published values triggers* SPRING *as the discrepancy distribution shifts. Both mechanisms operate on statistical signatures, not political categories.*

9 Specific Formal Applications

9.1 Fertility Rate Auditing

The total fertility rate (TFR) θ_{fert} is a critical governance metric with major implications for pension systems, education planning, and labor force projections. Three independent data sources provide cross-validation:

- **Census data A_1:** Decennial or quinquennial full-population enumeration. Provides the most comprehensive estimate but with long inter-census intervals. Standard error $\sigma_1 \approx 0.05$ births per woman.
- **Hospital birth records A_2:** Administrative data from birth registrations at hospitals and health facilities. Near-complete coverage in urban areas; may miss home births in rural regions. Standard error $\sigma_2 \approx 0.08$, with systematic downward bias in regions with low institutional delivery rates.
- **School enrollment A_3:** Primary school enrollment data lagged by 6–7 years provides an independent cohort-size estimate that can be back-projected to TFR using mortality and migration models. Standard error $\sigma_3 \approx 0.12$, with model-dependent uncertainty.
- **UN Population Division estimates A_4:** Bayesian hierarchical model combining all available national data with demographic transition priors. Standard error $\sigma_4 \approx 0.10$ for countries with high-quality data; larger for data-sparse settings.

The Yajie consensus TFR is:

$$\hat{\theta}_{\text{fert}}^{\text{Yajie}} = \sum_{j=1}^{4} w_j^{\text{Yajie}} \hat{\theta}_{\text{fert}}^{(j)}, \tag{72}$$

with the consensus standard error providing an auditable bound on TFR uncertainty. A published TFR m_{fert} is flagged if $\left| m_{\text{fert}} - \hat{\theta}_{\text{fert}}^{\text{Yajie}} \right| > 2\hat{\sigma}_{\text{Yajie}}$.

9.2 Employment Rate Auditing

Employment θ_{emp} exemplifies the power of multi-source verification. Four structurally independent data sources exist:

- **Household labor force survey A_1:** Stratified random sample of households, ILO-standard employment definitions. Standard error $\sigma_1 \approx 0.3$–0.5 percentage points for national estimates.
- **Payroll employment data A_2:** Census of formal-sector establishments reporting to tax/social security authorities. Covers formal employment only; misses informal sector, agricultural labor, and self-employment. Bias correlated with informality rate.
- **Social security contribution records A_3:** Administrative count of workers with active social security contributions. Similar coverage to payroll data but with different administrative incentives (workers may contribute without formal payroll).
- **Satellite nighttime lights A_4:** Machine learning model mapping VIIRS/DMSP nighttime light intensity to economic activity, calibrated to employment in surveyed regions. Standard error $\sigma_4 \approx 1.0$–2.0 percentage points at fine spatial resolution; improves with aggregation.

The structural independence of these sources is high: $\bar{\rho}$ between satellite and survey estimates is typically < 0.3, yielding $M_{\text{eff}} \approx M/(1 + (M-1) \cdot 0.3) \approx 2.1$ for $M = 4$. This is sufficient for opacity detection (Theorem 4.3) but may require additional auditors for transparency dominance (Theorem 3.2) depending on κ and L_B.

9.3 Government Openness Index

We formalize a **government openness index** as a directly measurable quantity from the publication record:

Definition 9.1 (Openness Index).

$$\Omega_t = \frac{K_{\mathrm{pub},t}}{K_{\min}} \cdot \frac{1}{3} \left(\underbrace{\frac{g_{\mathrm{pub},t}}{g_{\max}}}_{\textit{granularity}} + \underbrace{\frac{1}{1+\bar{\tau}_{\mathrm{lag},t}}}_{\textit{timeliness}} + \underbrace{\frac{f_{\mathrm{update},t}}{f_{\max}}}_{\textit{frequency}} \right), \tag{73}$$

where:

- $K_{\mathrm{pub},t}/K_{\min}$*: fraction of standard statistics published (from Theorem 4.3);*
- $g_{\mathrm{pub},t}/g_{\max}$*: average granularity of published data (e.g., provincial vs. county vs. township level), normalized by the maximum feasible granularity;*
- $\bar{\tau}_{\mathrm{lag},t}$*: average publication lag in months (lower is better);*
- $f_{\mathrm{update},t}/f_{\max}$*: update frequency (annual, quarterly, monthly, weekly), normalized.*

$\Omega_t \in [0,1]$ is directly verifiable: any observer can count published datasets, check their granularity, note the publication date, and track update frequency. It requires no statistical modeling—it is a *counting measure*. The SPRING mechanism monitors Ω_t for declines that may signal a transition toward information control.

9.4 Policy Evaluation: Causal Identification Strategies

For policy evaluation, the multi-expert framework employs causally distinct identification strategies:

(1) **Difference-in-Differences (DiD)** Compares outcome trends in treated vs. untreated units before and after policy implementation. Identification assumption: parallel trends in the absence of treatment.

(2) **Synthetic Control (SC)** Constructs a weighted combination of untreated units that matches the treated unit's pre-treatment trajectory. Identification assumption: the synthetic control approximates the counterfactual.

(3) **Regression Discontinuity Design (RDD)** Exploits a threshold in policy assignment (e.g., population cutoff for program eligibility). Identification assumption: units just above and below the threshold are comparable.

(4) **Instrumental Variables (IV)** Uses an exogenous source of variation in policy exposure. Identification assumption: exclusion restriction (instrument affects outcome only through policy exposure).

(5) **Structural estimation** Specifies and estimates a fully parametric economic model. Identification assumption: the model structure correctly captures all relevant mechanisms.

Each strategy has different identification assumptions. When they agree on the policy effect, the consensus is robust to assumption violations (the assumptions are logically independent). When they disagree, the pattern of disagreement reveals which assumptions are likely violated—exactly the logic of the unidentifiability theorem applied constructively.

10 M-Parameter as Management KPI MKPI

The M-parameter framework, when combined with cryptographic hashing (see the SCX Science Audit Mandate), transforms organizational accountability. This section formalizes M as a **management KPI** — a verifiable, non-gameable metric for individual and institutional trustworthiness.

10.1 The Accountability Chain

Definition 10.1 (M-Value Accountability Chain)**.** *An organization with L hierarchical layers. Each layer ℓ aggregates data from layer $\ell - 1$ and reports upward. Each report includes $(M_\ell, \mathcal{H}_\ell)$ where $\mathcal{H}_\ell = \text{SHA-256}(\text{data} \| M_\ell)$. Under mandatory M-declaration:*

(i) Worker submits: $(M_0, \mathcal{H}_0)$ — raw data with M-certification.

(ii) Manager aggregates: $(M_1, \mathcal{H}_1)$ — combines subordinate M-values, does not modify raw data.

(iii) Each subsequent layer ℓ passes upward: $(M_\ell, \mathcal{H}_\ell)$.

(iv) Public registry receives the chain $\{(M_\ell, \mathcal{H}_\ell)\}_{\ell=0}^{L}$.

Theorem 10.2 (Precise Blame Attribution (— Blame Shedding Theorem))**.** ■ ***[Full Proof]*** *When SCX detects $M_{obs} < M_{declared}$ anywhere in the chain, the* ***earliest layer*** *ℓ^* where the discrepancy first appears is uniquely identifiable: $\ell^* = \min\{\ell : \mathcal{H}_\ell$ inconsistent with the data at layer $\ell\}$. The responsible party at layer ℓ^* cannot shift blame upward (the hash at layer $\ell^* + 1$ merely transmitted their fraudulent input) nor downward (subordinates' hashes are independently verifiable).*

Proof. By induction on the chain. Each layer's hash $\mathcal{H}_\ell$ commits to the exact data received from layer $\ell - 1$. If $\mathcal{H}_0, \ldots, \mathcal{H}_{\ell-1}$ all verify and $\mathcal{H}_\ell$ does not, the discrepancy originated at layer ℓ. The fraudster at ℓ cannot claim the data came from $\ell - 1$ (hash mismatch) nor that $\ell + 1$ modified it (the hash at $\ell + 1$ was computed from the fraudulent input they provided). SHA-256 preimage resistance prevents post-hoc fabrication of alternative (M', data') matching $\mathcal{H}_\ell$. □ □

Corollary 10.3 (Perfect Blame Shedding for Honest Managers)**.** *A manager at layer ℓ who faithfully transmits subordinate M-values and hashes incurs zero liability for subordinate fraud. The fraud is attributed to layer $\ell^* < \ell$, and the manager's own M-value remains VERIFIED. This creates a* ***positive incentive*** *for managers to enforce M-declaration: not enforcing it makes them complicit (Theorem 4, Science Audit Mandate); enforcing it makes them immune.*

10.2 Game-Theoretic Analysis

Proposition 10.4 (M-KPI as a Dominant Strategy Enforcer)**.** *Under the M-accountability chain with public registry:*

(i) ***Workers****: Honest declaration ($M = M_{true}$) strictly dominates any falsification strategy when the detection probability $\mathbb{P}(\text{detection}) \to 1$ (Theorem 3, Science Audit Mandate).*

(ii) ***Managers****: Requiring M-declaration from subordinates strictly dominates not requiring it — the former provides blame immunity (Corollary above); the latter makes the manager complicit (Theorem 4, Science Audit Mandate).*

(iii) ***Organization****: Publishing the full M-chain on the public registry strictly dominates selective publication — incomplete chains are equivalent to $M = 0$ by Theorem 2 (Science Audit Mandate).*

The Nash equilibrium is universal honest M-declaration at all layers.

10.3 Practical Implications

The M-KPI transforms organizational dynamics:

- **Whistleblowing becomes unnecessary**. The M-chain automatically exposes fraud; individual workers need not risk retaliation — the mathematics does the exposure.
- **Internal investigations become trivial**. When $M_{\text{obs}} < M_{\text{declared}}$, the hash chain identifies the exact layer and individual responsible. No need for committees, hearings, or testimony.
- **Performance evaluation gains an objective dimension**. Employee M-values are auditable quality metrics. High M = verifiable competence. Low M = objectively poor data practices.
- **Organizational reputation becomes quantifiable**. An organization's aggregate M-value is publicly visible. Investors, regulators, and partners can compare M-values across organizations.

Remark 10.5 (limitation)**.** *The M-KPI does not measure effort, creativity, or qualitative contribution. It measures* ***data integrity****. Organizations should use M as a floor (minimum acceptable data quality), not a ceiling (substitute for holistic evaluation).*

10.4 Relationship to Existing Theoretical Frameworks

Mechanism Design (Myerson). Our framework is a mechanism design problem: we construct a game (the Yajie audit game) such that the equilibrium strategy (honest reporting) achieves a social objective (accurate governance statistics). The mechanism design literature [4, 5] establishes that truthful revelation requires incentive compatibility. The Yajie payoff structure achieves incentive compatibility through the detection penalty $\kappa \cdot \mathbb{P}(\|m - c\| > \varepsilon)$, which makes the expected cost of misreporting exceed the benefit when $M > M^*$. Unlike classical mechanism design, we do not require a central planner to enforce penalties; the auditor community provides decentralized enforcement through its consensus estimates. However, this requires that detection carries real consequences ($\kappa > 0$), which in turn requires that the auditor community's findings influence the government's payoff (through reputation, market access, institutional sanctions, or domestic accountability mechanisms). The mechanism is *incomplete* if $\kappa = 0$—a fundamental limitation we discuss below. **Transparency Theory (Stiglitz).** Stiglitz's work on information asymmetry [6, 7] establishes that transparency improves market efficiency by reducing information rents. Our framework extends this logic to governance statistics: the auditor community reduces the government's information monopoly, diminishing its ability to extract "governance rents" from favorable misreporting. Theorem 3.2 provides a quantitative formalization: the threshold M^* is the point at which the information rent ($L_B\delta$) is eliminated by detection risk. The novelty is in the explicit role of *multiplicity*: transparency is not a binary property but a continuous function of the number and independence of auditors. **Social Choice (Arrow).** Arrow's impossibility theorem [8] establishes that no social welfare function can simultaneously satisfy a set of seemingly reasonable axioms. Our framework sidesteps this impossibility by restricting its scope: we do not aggregate preferences or rank social states. We verify statistical claims against multi-expert consensus. The Yajie consensus is a statistical aggregation of estimates about an objective quantity θ, not a preference aggregation. This is a crucial distinction: governance statistics are (in principle) verifiable facts, not value judgments. The SCX framework audits the former, not the latter.

10.5 Honest Limitations

We now state what SCX governance auditing cannot do, consistent with the SCX mandate of honest limitation declaration.

[L1] **[L1] Cannot replace democratic deliberation .** SCX detects statistical anomalies in published governance data. It cannot determine whether a particular unemployment rate is "too high," whether a fertility policy is "just," or whether a tradeoff between growth and equality is "acceptable." These are normative questions requiring democratic processes. The mathematics tells us what *is* (or what is likely to be true given the data); it does not tell us what *should be*.

[L2] **[L2] Cannot resolve value conflicts .** When experts disagree about policy effects, the Yajie consensus produces a weighted average with confidence bounds. It does not adjudicate which expert's normative framework is superior. A Marxist economist and a neoclassical economist may produce different policy effect estimates because they model different mechanisms; the consensus averages their numbers but does not resolve their theoretical disagreement.

[L3] **[L3] Cannot detect shared biases .** If all auditors share a common systematic bias (Assumption 2.13 violated), the consensus inherits that bias. For example, if all employment estimates rely on the same flawed population denominator, the consensus will be precisely wrong. SCX detection power depends on *diversity* of data sources and methodologies; without it, the audit is blind.

[L4] **[L4] Requires $\kappa > 0$ for incentive compatibility $\kappa > 0$.** The transparency dominance of Theorem 3.2 requires that detection carries a real cost. If the government is indifferent to auditor findings ($\kappa = 0$), the audit is a measurement exercise with no behavioral effect. This is not a mathematical limitation but an institutional one: audit without consequences is documentation, not governance.

[L5] **[L5] Cannot verify counterfactuals .** Policy evaluation (Section 5) requires estimating what would have happened without the policy—an inherently unverifiable counterfactual. The multi-expert consensus bounds the uncertainty but cannot eliminate it. Causal inference from observational data always requires untestable assumptions [9].

[L6] **[L6] Auditor independence is asymptotic .** Assumption 2.11 (conditional independence) is an idealization. In practice, auditors share methods, data sources, and professional networks, inducing positive correlation $\bar{\rho} > 0$. The effective multiplicity correction M_{eff} partially addresses this, but the independence assumption should be treated as a target for institutional design, not a description of current reality.

[L7] **[L7] Cannot substitute for institutional legitimacy .** A statistical audit, however rigorous, does not confer democratic legitimacy. A government that achieves perfect Cercis scores on statistical accuracy may still lack popular mandate. SCX audits the *output* of governance (data quality), not the *process* (elections, representation, accountability). Both are necessary; neither is sufficient.

[L8] **[L8] Regime detection has latency .** The Spring mechanism (Section 8) detects regime transitions with expected delay $O(\log(1/\delta)/(1-\lambda))$. For gradual transitions (small δ), this delay can be substantial. Rapid detection requires high-frequency data and low λ, which increases false alarm rates. There is an unavoidable speed-accuracy tradeoff.

10.6 Future Directions

(i) **Empirical calibration of M^*.** The threshold auditor multiplicity M^* depends on parameters (κ, L_B, $\bar{\sigma}^2$, $\bar{\rho}$) that must be estimated from historical governance data. A systematic empirical study calibrating M^* for different countries and governance metrics would operationalize the framework.

(ii) **Dynamic audit games.** Our model treats the audit as a one-shot game. In reality, governments and auditors interact repeatedly, creating reputation effects, learning, and strategic adaptation. A repeated-game extension would capture these dynamics.

(iii) **Hierarchical auditor communities.** Our model treats all auditors as symmetric. In practice, auditors have different credibility, resources, and access. A hierarchical YAJIE consensus with auditor-level quality weights (themselves subject to meta-audit) would improve robustness.

(iv) **Integration with causal machine learning.** Modern causal ML methods (double/debiased machine learning, causal forests, deep instrumental variables) can serve as additional experts in the multi-expert framework, increasing M and reducing $\bar{\rho}$ through methodological diversity.

(v) **Institutional design implications.** The theorems provide quantitative guidance for designing transparency institutions: Theorem 3.2 tells us how many independent statistical agencies are needed; Theorem 4.3 tells us the detection probability as a function of publication completeness.

11 Conclusion Conclusion

We have presented a mathematical framework for SCX auditing of governance, grounded in game theory, statistical detection theory, and causal identification. The framework formalizes governance as a signaling game with multi-expert audit under the YAJIE Nash-Pareto Equilibrium payoff structure. Three core theorems establish: (i) transparency dominance—under sufficient auditor multiplicity, honest reporting is the game-theoretic equilibrium (Theorem 3.2); (ii) opacity detection—withholding statistics is probabilistically detectable, making full publication the best response (Theorem 4.3); and (iii) policy unidentifiability—deviation from predicted policy outcomes cannot be attributed to a specific cause without declared assumptions (Theorem 5.3). The framework is deliberately mathematical, not normative. We do not argue that transparency is "good" or that audit is "desirable." We prove that under specified conditions, transparency emerges as the rational strategy. The conditions are explicit: sufficient auditor multiplicity ($M > M^*$), meaningful detection penalties ($\kappa > L_B\delta_{\min}$), low auditor error correlation ($\bar{\rho} \ll 1$), and published statistics completeness ($K_{\text{pub}} \to K_{\min}$). When these conditions fail, the theorems predict that misreporting and opacity may be rational strategies—and the mathematics identifies exactly what structural changes would shift the equilibrium. The SCX framework's contribution to governance is not a political program but a **verification technology**: a set of mathematical tools for assessing the quality of published governance statistics through multi-expert consensus. Like any technology, it can be used well or poorly, adopted or ignored. Its value lies in making the transparency problem computationally tractable and its assumptions explicit. **Acknowledgments.** We thank the SCX for the foundational framework.

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